Word Level Autoregressive Limerick Generation System

Foundations of Generative AI ICT4403 Assignment 2

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Assignment title	Foundations of Generative AI Assignment 2
System title	Word Level Limerick Generation with RNN LSTM and GRU
Dataset	Limerick Dataset OEDILF v3 - https://doi.org/10.5281/zenodo.5722527
GitHub repository	https://github.com/mizar-0/RNN-LSTM-GRU-Comparison

Executive summary. Four word-level autoregressive language models were trained on the full filtered OEDILF limerick corpus under comparable settings. The GRU achieved the strongest held-out next-word prediction performance and the best combined generation evidence. The recommended configuration is the GRU sampled at temperature 0.3.

1 Problem and objective

The task was to develop an autoregressive text-generation system using Simple RNN, Bidirectional RNN, LSTM, and GRU architectures. Each model predicts the next word from its preceding context and generates text sequentially from the same seed sequences. The objectives were to compare predictive performance, generated-text quality, and the effect of sampling temperature.

2 Dataset and preprocessing

The Limerick Dataset OEDILF v3 was obtained from Zenodo under the CC BY 3.0 US licence. The source file contains 98,454 records, of which 65,886 were labelled as limericks and used in the final experiment. The data were split by poem into 80% training, 10% validation, and 10% test partitions, preventing token windows from the same poem appearing in multiple splits.

Text was lowercased and tokenised into words and punctuation. Each poem received beginning-of-sequence, end-of-line, and end-of-sequence tokens. A vocabulary of up to 10,000 training-set tokens was created, with pad and unknown tokens retained. Each target token was paired with a 64-token left context, creating next-word input-target examples.

3 Models and experimental protocol

The Simple RNN, Bidirectional RNN, LSTM, and GRU each used 128-dimensional embeddings, 256 hidden units, dropout 0.3, Adam optimisation with learning rate 0.001, batch size 512, a 64-token context, and random seed 42. Training was capped at 40 epochs with validation-loss early stopping (patience 5). The best validation checkpoint for each model was evaluated once on the held-out test set.

4 Quantitative comparison

Model	Epochs	Parameters	Test PPL	Top-1	Top-5	Time
Simple RNN	22	3.95M	62.79	31.94%	50.56%	21.8 min
BiRNN	33	6.61M	62.81	32.00%	50.61%	51.7 min
LSTM	26	4.25M	50.03	34.14%	53.55%	29.9 min
GRU	40	4.15M	45.30	34.84%	54.79%	47.0 min

The gated models substantially outperformed the vanilla recurrent models. GRU obtained the lowest test perplexity (45.30) and highest top-1 and top-5 accuracies. BiRNN used the most parameters and the most time but was effectively tied with the smaller Simple RNN on test perplexity.

5 Temperature study and qualitative evaluation

A reproducible sweep generated 7,200 poems: four models, six temperatures (0.3, 0.5, 0.8, 1.0, 1.3, and 1.5), and 300 poems per model-temperature cell. All models used the same five seed prompts. Automatic evaluation measured termination, exact five-line form, CMU Pronouncing Dictionary coverage, an AABBA rhyme score, repeated trigrams, immediate word repetitions, copied training lines, distinct n-grams, and unique-poem share. Bootstrap percentile intervals used 2,000 resamples.

The expected trade-off appeared clearly. Increasing temperature increased lexical diversity but reduced adherence to limerick form and rhyme. At temperature 0.3, GRU generated exactly five lines for 100% of samples and obtained the strongest AABBA rhyme score (46.7%). The held-out human reference row had an approximately 99.9% rhyme score, showing that all neural models remained substantially below the source-text form quality.

6 Blinded manual scoring

A blinded manual sample contained 80 poems: four models, four temperatures, and five poems per cell. Poems were rated from 1 to 5 for coherence and contextual consistency without revealing the model or temperature. GRU at temperature 0.3 had the strongest combined manual score (3.0/5, 95% bootstrap CI 2.6-3.4), tied in point estimate with LSTM at temperature 0.3. GRU had better contextual consistency (3.4/5) and much stronger automatic rhyme evidence. At high temperature, manual scores deteriorated for every model.

7 Conclusion

GRU at temperature 0.3 is the most effective configuration for this dataset. It combines the best next-word prediction metrics, reliable five-line termination, the highest automatic rhyme adherence, and the strongest contextual-consistency rating. Lower temperature is preferable for structured limerick generation because it preserves form, although it reduces diversity.

8 Limitations

The study used one random seed, early stopping yielded different epoch counts across models, and GRU was still improving at the epoch cap. The rhyme metric only applies when line-ending words are covered by the CMU Pronouncing Dictionary. Manual scores used one rater and five poems per cell, so overlapping intervals should not be treated as precise rank differences. Perplexity measures next-token prediction and does not fully capture poetic coherence or humour.

References

Limerick Dataset OEDILF v3. Zenodo. https://doi.org/10.5281/zenodo.5722527